VELLORE INSTITUTE OF TECHNOLOGY

SCOPE | Department of IoT — BACSE202: Database Systems

Course Project

Interfacing the Database with a Front End using ODBC / JDBC

A 3-Phase Project • Assessments 8–10 • CO5 • 30 Marks • Teams of 3

Extended from Lab Experiment 9 (Front-End Database Connectivity)

Objectives

1

Design a complete database application

Move from requirement analysis to a fully working, front-end-connected system.

2

Apply core DB theory

ER/EER modelling, relational mapping, and normalization on a real-world problem.

3

Integrate connectivity concepts

Combine database design with front-end and business logic via ODBC/JDBC.

4

Demonstrate incremental progress

Three structured phases, each independently evaluated (CO5).

Project Overview

Lab Experiment 9, extended into a semester-long team project

What it is

- Extends Lab 9: “Interface the database with Front Ends using ODBC/JDBC; perform retrieval and DML operations from the Front End.”
- Executed as ONE continuous application, evaluated across 3 phases.
- Each phase = a separate assessment (CO5), building on the last.

At a Glance

3
Phases

3
Students / Team

30
Total Marks

CO5
Course Outcome

Assessment & Timeline

Assessment	Phase	A1 Slot	A2 Slot	Completion	Marks	Submission
Assessment 8	Phase 1	05-Oct-2026	08-Oct-2026	30%	10 M	Phase 1 PPT
Assessment 9	Phase 2	12-Oct-2026	15-Oct-2026	80%	10 M	Phase 2 PPT
Assessment 10	Phase 3	26-Oct-2026	29-Oct-2026	100%	10 M	Final PPT + Project Report

Total: 30 marks under CO5. Every phase must demonstrate live ODBC/JDBC connectivity.

Project Stages Mapped to Phases

Eleven execution stages, distributed across three progressive phases

Phase 1 — 30%	Phase 2 — 80%	Phase 3 — 100%
Problem Identification & Requirement Analysis	Database Design & Table Creation (DDL)	Database Connectivity (complete)
ER / EER Diagram Construction	Data Population (DML)	Testing and Validation
ER Diagram to Relational Model	Front-End Design	Demonstration & Documentation
Normalization (1NF, 2NF, 3NF)	Business Logic Implementation	
	Database Connectivity (initial)	

Stage-wise Mark Rubric (CO5)

Suggested evaluation framework — 11 stages distributed across 3 phases, 30 marks total

#	Stage	Phase	Marks
1	Problem Identification and Requirement Analysis	**Phase 1**	**2**
2	Construction of ER/EER Diagram	**Phase 1**	**3**
3	ER Diagram to Relational Model	**Phase 1**	**3**
4	Normalization (1NF, 2NF, 3NF)	**Phase 1**	**2**
5	Database Design and Table Creation (DDL)	**Phase 2**	**2**
6	Data Population (DML)	**Phase 2**	**2**
7	Front-End Design	**Phase 2**	**2**
8	Business Logic Implementation	**Phase 2**	**2**
9	Database Connectivity	**Phase 2 → 3**	**2 + 2**
10	Testing and Validation	**Phase 3**	**3**
11	Project Demonstration and Documentation	**Phase 3**	**5**

Note: Database Connectivity (Stage 9) is assessed twice — an initial working connection in Phase 2 (2 marks) and full CRUD integration in Phase 3 (2 marks) — giving each phase a 10-mark total (30 marks overall).

Phase 1 — Design Foundation

Assessment 8 • 30% Completion • 10 Marks • Submission: Phase 1 PPT

Deliverables

- Submission format: a single Phase 1 PPT (via VTOP) covering all stages below
- Problem statement & requirement analysis (scope, users, functional requirements)
- ER/EER diagram (draw.io) with entities, attributes, relationships, keys, cardinalities
- Mapping of the ER/EER diagram to a relational schema
- Normalization worksheet: 1NF → 2NF → 3NF with justification

Rubric — 10 Marks

Criteria	Marks
1. Problem Identification & Requirement Analysis	**2**
2. Construction of ER/EER Diagram	**3**
3. ER Diagram to Relational Model	**3**
4. Normalization (1NF, 2NF, 3NF)	**2**

Phase 2 — Implementation Core

Assessment 9 • 80% Completion • 10 Marks • Submission: Phase 2 PPT

Deliverables

- Submission format: a single Phase 2 PPT (via VTOP) covering all stages below
- DDL scripts: all tables with constraints, consistent with the Phase 1 schema
- DML scripts populating tables with meaningful sample data
- Working front-end interface for the application
- Core business logic behind the front end (validations, workflow rules)
- Initial ODBC/JDBC connection performing basic retrieval & DML from the front end

Rubric — 10 Marks

Criteria	Marks
5. Database Design & Table Creation (DDL)	**2**
6. Data Population (DML)	**2**
7. Front-End Design	**2**
8. Business Logic Implementation	**2**

Phase 3 — Integration & Delivery

Assessment 10 • 100% Completion • 10 Marks • Submission: Final PPT + Project Report

Deliverables

- Submission format: Final PPT + complete Project Report (via VTOP)
- Complete ODBC/JDBC integration supporting all CRUD operations
- Test cases: functional correctness, constraint violations, error handling
- Fully functional application matching 100% of proposed scope
- Project report: schema, scripts, screenshots, test case results, end-to-end summary
- Live project demonstration and viva

Rubric — 10 Marks

Criteria	Marks
9. Database Connectivity (complete)	**2**
10. Testing and Validation	**3**
11. Demonstration and Documentation	**5**

Team Formation

3 students per project — roles may rotate across phases

1

Database Designer

ER/EER modelling, normalization, DDL scripts.

2

Front-End & Business Logic Developer

UI design, workflow rules, validations.

3

Integration, Testing & Documentation Lead

ODBC/JDBC connectivity, test cases, project report.

Every member must understand the complete project — individual viva is conducted at each phase.

Do's and Don'ts

Do's

- Finalize ER/EER & normalized schema in Phase 1 before writing DDL
- Keep schema and front end consistent with the Phase 1 problem statement
- Test each SQL operation in SQL*Plus before wiring it to the front end
- Maintain one coherent application scope across all 3 phases
- Keep each phase PPT self-contained, but consistent with earlier phases

Don'ts

- Don't leave connectivity for Phase 3 — start it in Phase 2
- Don't submit a schema not normalized through 1NF–3NF
- Don't hardcode data in the front end
- Don't change team composition after Phase 1 without approval

Submission Guidelines

Phase 1 & 2 — PPT	Submit a structured Phase 1 / Phase 2 PPT via VTOP, covering the stages assigned to that phase (problem/design for Phase 1; build/connect for Phase 2).
Phase 3 — Final PPT + Report	Submit a Final PPT together with a complete Project Report (end-to-end: design, scripts, screenshots, test results) via VTOP.
One submission per team	Single team submission per phase, listing all member names and registration numbers.
Live demo & viva	Each phase submission is accompanied by a live demonstration during the lab slot (A1 / A2 as scheduled).
Late policy	Late submissions follow the standard VIT academic policy for lab/project assessments.

Suggested Project Ideas

Ten ideas spanning the syllabus and current application trends

1. **Smart Hospital & IoT Vitals Monitoring**
   Healthcare + IoT
2. **Smart Campus Hostel & RFID Attendance**
   Smart Campus + IoT
3. **E-Commerce Order & Inventory Management**
   E-Commerce
4. **Online Library with Book Recommendation**
   EdTech
5. **Smart Agriculture & Sensor Data Management**
   AgriTech + IoT
6. **Online Examination & Result Analytics**
   EdTech
7. **Food Delivery & Restaurant Order Management**
   FoodTech
8. **Bank Loan, Account & Transaction Management**
   FinTech
9. **Fitness & Wearable Health Data Tracking**
   HealthTech + IoT
10. **Employee Payroll & HR Management**
    Enterprise

Build. Connect. Demonstrate.

Three phases, one working database application — designed, populated, connected, and validated.

Reference: Elmasri & Navathe, Fundamentals of Database Systems